

Supplementary Material

ABC-Conditioned Score Correction for Dermoscopic Melanoma Classification under External Dataset Shift

Supplementary Methods

All supplementary analyses used the same frozen prediction files and train/validation/test definitions described in the main manuscript. No classifier or image backbone was retrained for the supplementary analyses. Recalibration and ABC-conditioned score correction were fitted only on the HAM10000 validation split and evaluated without refitting on HAM10000 test and BCN20000.

S1. Split integrity and dataset provenance

Lesion-level split integrity was verified by checking overlap of `lesion_id` across train, validation, and test partitions. No lesion identifier crossed partitions.

Table 1: HAM10000 lesion-level split integrity.

Check	Train-validation	Train-test	Validation-test
Overlapping lesion identifiers	0	0	0

The local ISIC 2018 Task 3 files overlapped completely with HAM10000 and were therefore treated only as a secondary melanoma-label audit. BCN20000 was used as the independent external validation cohort after duplicate screening.

S2. ISIC 2018 label-audit curves

The local ISIC 2018 Task 3 files overlapped completely with HAM10000. These curves are therefore shown only as a label-audit analysis, not as external validation.

S3. DeLong ROC-AUC comparisons

Full pairwise DeLong comparisons are provided in the released CSV files: `table_delong_auc_tests.csv` and `table_abccal_delong_auc_tests.csv`.

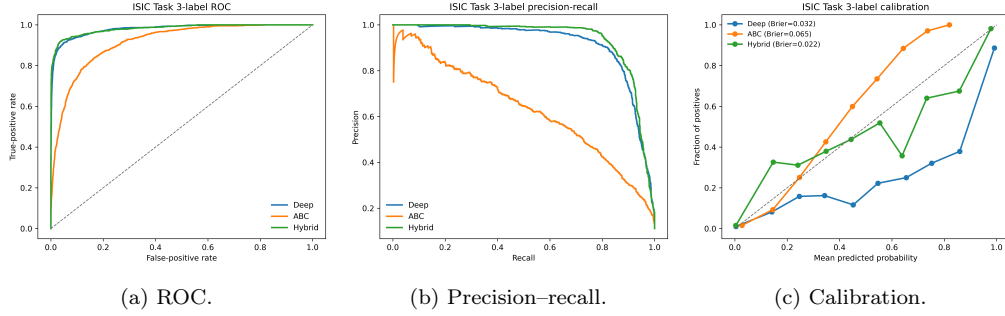


Figure 1: Secondary ISIC 2018 Task 3 melanoma-label audit curves on the overlapping HAM10000/ISIC image corpus.

Table 2: Key DeLong comparisons for main model families. Holm-adjusted p values are shown.

Dataset	Comparison	AUC difference	95% CI low	95% CI high	Holm p
HAM10000 test	Deep minus Hybrid	0.0026	-0.0110	0.0161	0.712
BCN20000	Hybrid minus Deep	0.0482	0.0405	0.0558	3.93×10^{-34}
BCN20000	Hybrid minus ABC-only	0.0692	0.0574	0.0810	8.29×10^{-30}
BCN20000	Deep minus ABC-only	0.0210	0.0073	0.0347	0.008

Table 3: Key DeLong comparisons for deep-score correction. Holm-adjusted p values are shown.

Dataset	Comparison	AUC difference	95% CI low	95% CI high	Holm p
HAM10000 test	Raw minus ABC-conditioned	0.0025	-0.0084	0.0133	1.000
BCN20000	ABC-conditioned minus Raw	0.0419	0.0335	0.0503	1.16×10^{-21}
BCN20000	ABC-conditioned minus Platt	0.0419	0.0335	0.0503	1.16×10^{-21}
BCN20000	ABC-conditioned minus Isotonic	0.0443	0.0359	0.0527	5.09×10^{-24}

S4. Calibration curves and bin-level summaries

Calibration curves with uncertainty intervals are shown for the main model families and for deep-score correction methods. Full bin-level summaries are provided in `table_calibration_curve_bins.csv` and `table_abccal_calibration_curve_bins.csv`.

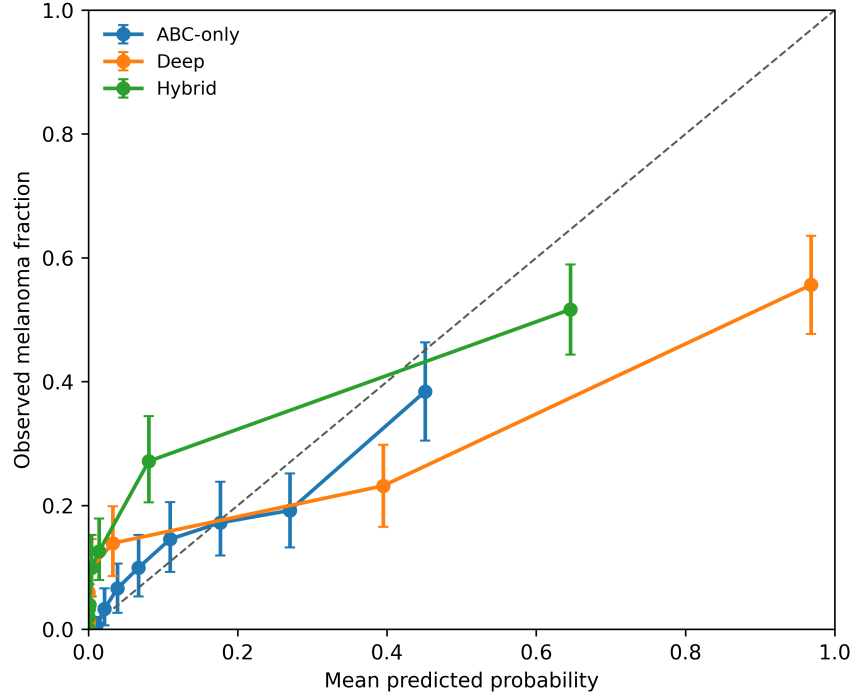


Figure 2: HAM10000 calibration curves for ABC-only, deep, and hybrid models with binomial uncertainty intervals.

S5. Subgroup calibration

Full subgroup performance and bootstrap calibration summaries are provided in `table_bcn_subgroup_performance.csv` and `table_subgroup_calibration_bootstrap.csv`. These analyses are descriptive transportability checks rather than formal fairness tests.

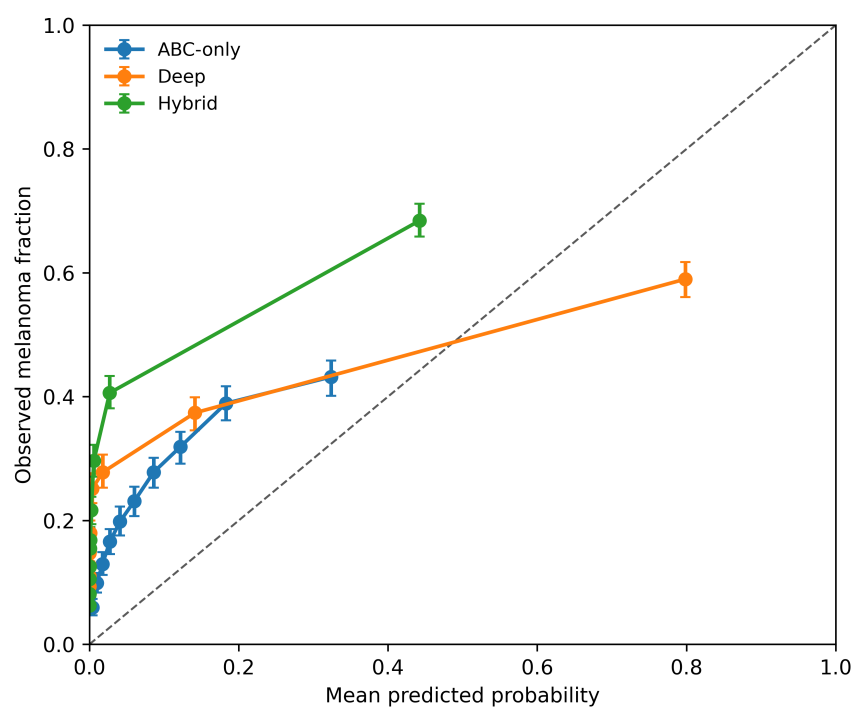


Figure 3: BCN20000 calibration curves for ABC-only, deep, and hybrid models with binomial uncertainty intervals.

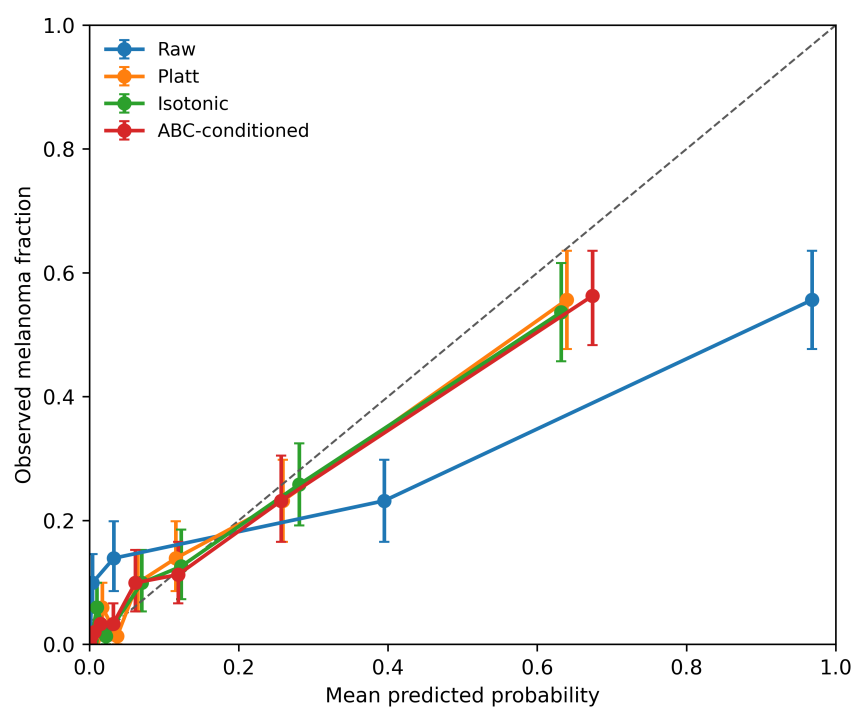


Figure 4: HAM10000 calibration curves for raw deep probabilities, Platt scaling, isotonic regression, and ABC-conditioned score correction.

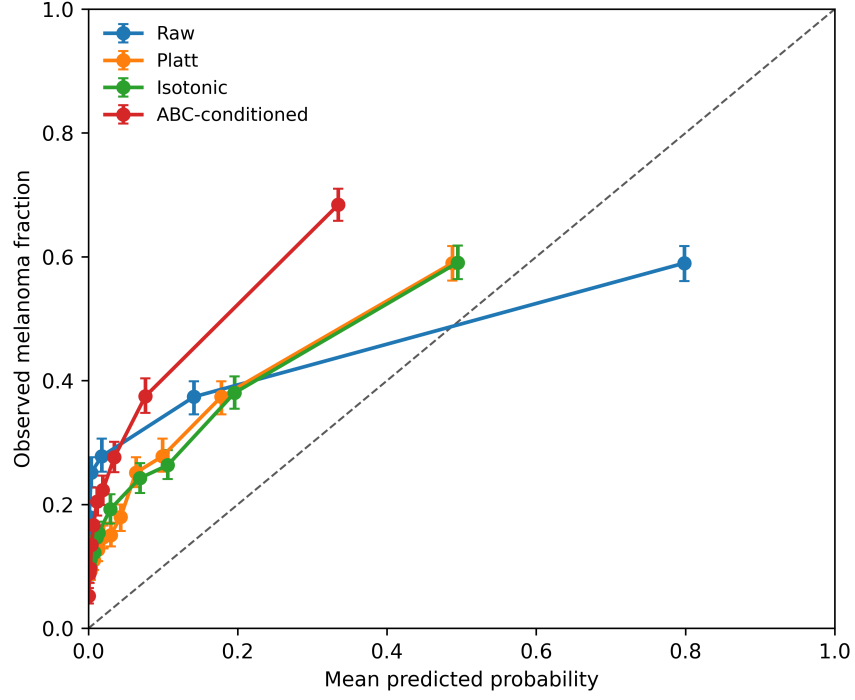


Figure 5: BCN20000 calibration curves for raw deep probabilities, Platt scaling, isotonic regression, and ABC-conditioned score correction.

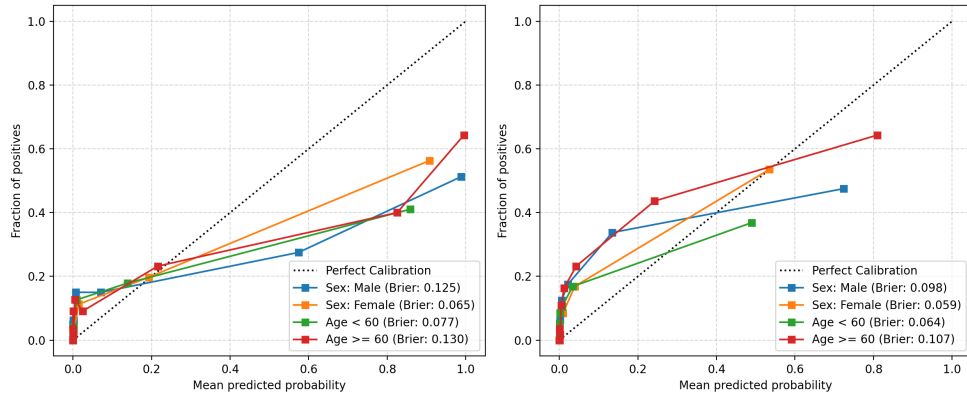


Figure 6: HAM10000 subgroup calibration by sex and age strata.

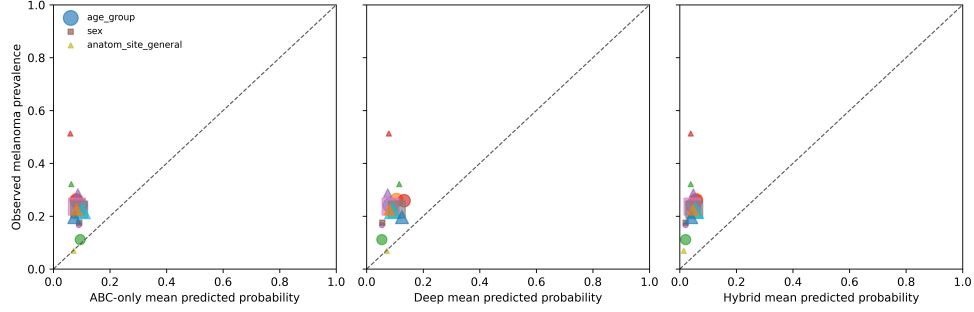


Figure 7: BCN20000 subgroup calibration by age group, sex, and anatomical site. Marker size reflects subgroup size.

S6. ABC-conditioned score-correction ablations

Full ABC-conditioned score-correction ablations are provided in `table_abccal_ablation.csv`. Bootstrap contrasts on BCN20000 are provided in `table_abccal_bootstrap_bcn.csv`. These analyses show that descriptor-conditioned correction affects discrimination, calibration, and decision-curve utility differently depending on whether slope, intercept, or descriptor-group-specific terms are used.

S7. Grad-CAM localization check

Grad-CAM heatmaps were compared with ISIC 2018 Task 1 lesion masks. The mean fraction of top 20% saliency inside the lesion mask was 0.154, with median 0.000. Mean saliency was higher outside than inside the lesion mask. These results indicate that Grad-CAM maps should be interpreted as failure-mode audits rather than localization validation.

S8. Segmentation-sensitivity analysis

Segmentation-sensitivity analysis was planned but could not be completed because Otsu-derived HAM10000 and BCN20000 feature files were not retained in the current pipeline version. The analysis was therefore skipped and no downstream segmentation-sensitivity result was fabricated. The skipped-analysis record is provided in `table_segmentation_sensitivity.csv`.

S9. CLAIM/TRIPOD+AI reporting map

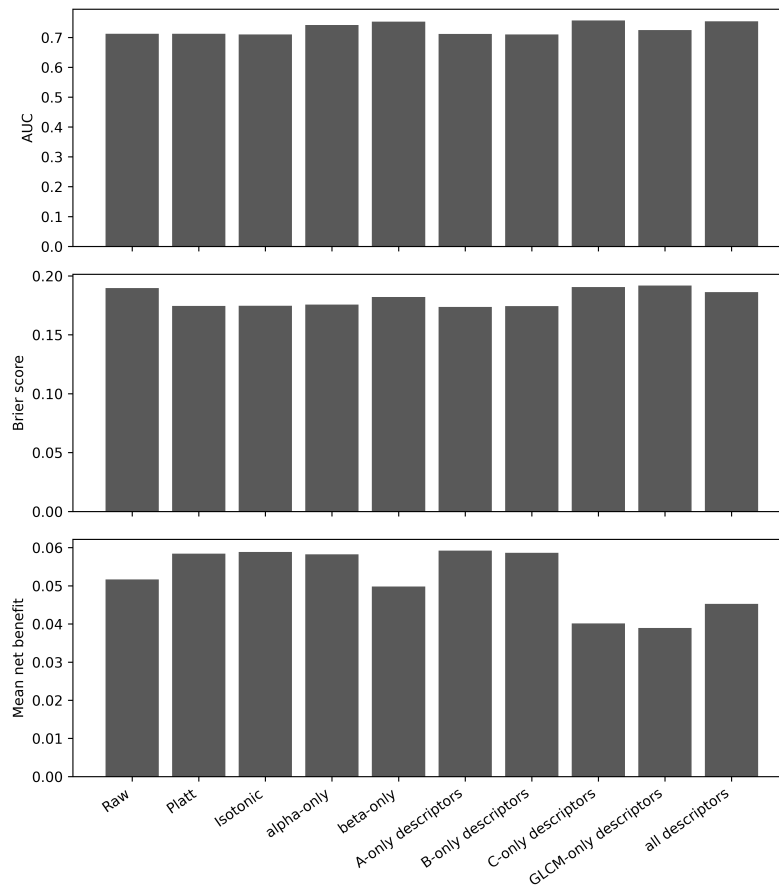


Figure 8: BCN20000 ablation of ABC-conditioned score correction across AUC, Brier score, and mean decision-curve net benefit.

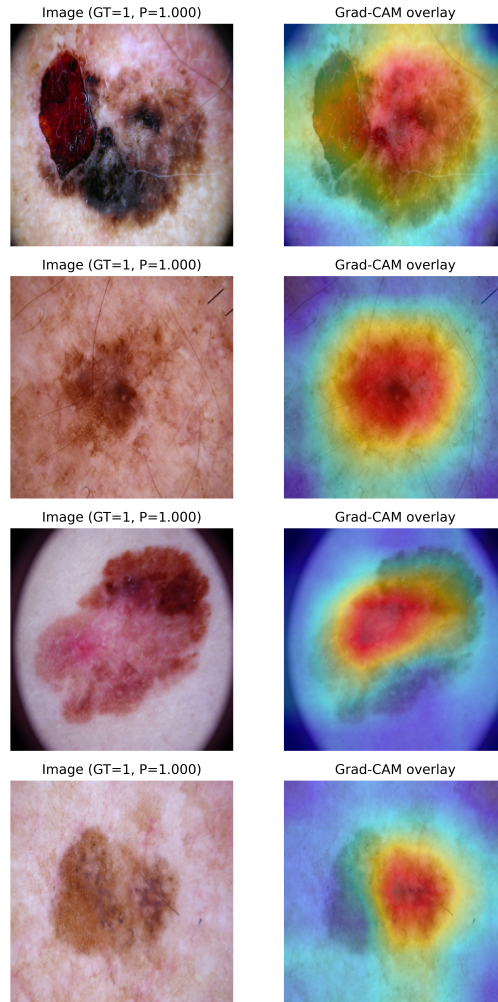


Figure 9: Qualitative Grad-CAM examples for selected deep-baseline cases. These examples are used only for visual failure-mode auditing and not as localization validation.

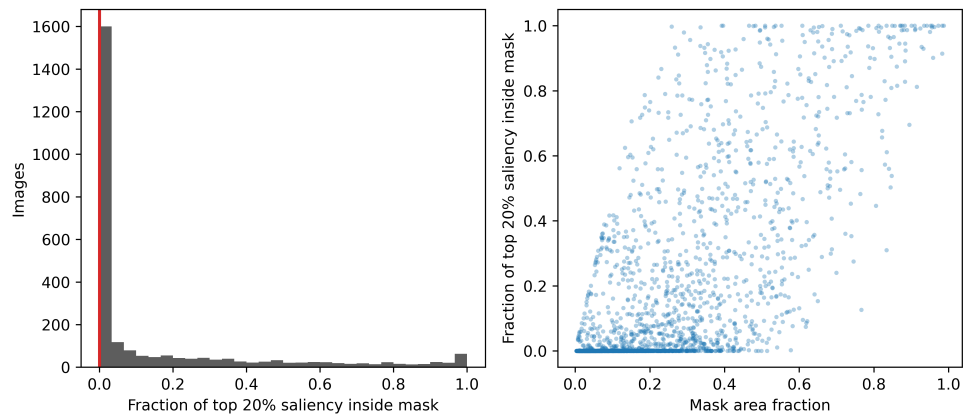


Figure 10: Grad-CAM localization check. The left panel shows the distribution of top-saliency overlap with lesion masks. The right panel shows the relationship between lesion area fraction and saliency overlap.

Table 4: Pragmatic reporting map for CLAIM and TRIPOD+AI items. The table groups related checklist items and indicates where they are addressed in the main manuscript or Supplementary Material.

Reporting domain	Where addressed	Comment
Title and abstract	Main manuscript: Title and Abstract	Identifies the dermoscopic melanoma task, external dataset shift, score correction, and AI model families.
Clinical or scientific rationale	Main manuscript: Introduction	Motivates separate assessment of discrimination, calibration, decision utility, dataset shift, and interpretability.
Intended use and users	Main manuscript: Introduction	States that the framework is for retrospective model evaluation and audit, not autonomous diagnosis or clinical deployment.
Datasets and study setting	Main manuscript: Materials and methods; dataset summary, provenance, and subset-prevalence tables	Describes HAM10000, ISIC 2018 Task 3 label audit, and BCN20000 external validation subset.
Eligibility, sample size, and class balance	Main manuscript: dataset summary, provenance, and subset-prevalence tables	Reports image counts, lesion counts where available, melanoma counts, non-melanoma counts, and prevalence.
Reference standard and endpoint definition	Main manuscript: Datasets and endpoint	Dataset-provided diagnosis labels were harmonized to melanoma versus all other diagnoses.
Data leakage and overlap control	Main manuscript: Datasets and endpoint; Supplementary Section S1; Supplementary Table 1	Reports lesion-level split integrity and duplicate screening of BCN20000 using identifiers and perceptual hashes.
Image preprocessing	Main manuscript: Preprocessing and segmentation	Reports resizing, ImageNet normalization, fixed random seed, and segmentation postprocessing.
Segmentation method and benchmarking	Main manuscript: Preprocessing and segmentation; segmentation-benchmark table; Supplementary Section S8	Benchmarks Otsu and level-set segmentation on ISIC 2018 Task 1 masks and notes the skipped downstream segmentation-sensitivity analysis.
Predictors and feature extraction	Main manuscript: Features and models; Score correction	Describes ABC descriptors, EfficientNet-B0 embeddings, and descriptor inputs to ABC-conditioned score correction.
Model architecture and training	Main manuscript: Features and models; model-parameter table	Reports EfficientNet-B0, XGBoost, optimizer, learning rate, epochs, class weighting, and model-selection criterion.
Internal validation	Main manuscript: Evaluation; Results: HAM10000 test	Uses lesion-level HAM10000 train/validation/test separation with held-out test evaluation.
External validation/testing	Main manuscript: Datasets and endpoint; Results: BCN20000 validation; external-validation table and curves	Frozen HAM10000-trained models were evaluated on the duplicate-screened BCN20000 subset without refitting.

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Table 4: Pragmatic reporting map for CLAIM and TRIPOD+AI items (continued).

Reporting domain	Where addressed	Comment
Performance metrics	Main manuscript: Evaluation; Results: HAM10000 test and BCN20000 validation	Reports ROC-AUC, PR-AUC, Brier score, F1, sensitivity, specificity, and accuracy at threshold 0.5.
Statistical uncertainty and comparisons	Main manuscript: Statistical analysis; Supplementary Section S3; Supplementary Tables 2 and 3	Reports DeLong tests, Holm correction, lesion-clustered bootstrap, and bootstrap calibration contrasts.
Calibration and probability reliability	Main manuscript: Evaluation, Statistical analysis, and Score correction; Supplementary Section S4	Reports Brier score, ECE, calibration curves, calibration slope/intercept, Platt scaling, isotonic regression, and ABC-conditioned score correction.
Decision-curve analysis	Main manuscript: Evaluation; Results: BCN20000 validation and Score correction	Evaluates threshold-dependent net benefit over threshold probabilities 0.02–0.50.
Dataset shift and transportability	Main manuscript: Statistical analysis; Results: Dataset shift	Reports descriptor SMDs, Kolmogorov–Smirnov statistics, embedding-space PCA, and score-distribution shift.
Model updating or post-hoc correction	Main manuscript: Score correction; Results: Score correction and decision curves; Supplementary Section S6	Describes validation-fitted Platt, isotonic, and ABC-conditioned correction, then evaluates them without BCN20000 refitting.
Explainability and interpretability	Main manuscript: Evaluation; Concept probes; Feature and saliency analysis	Includes concept probes, SHAP-style feature contributions, and Grad-CAM saliency checks.
Subgroup and heterogeneity checks	Main manuscript: Concept probes and subgroups; Supplementary Section S5	Reports descriptive subgroup calibration and performance checks; not interpreted as formal fairness testing.
Failure-mode analysis	Main manuscript: Feature and saliency analysis; Supplementary Section S7	Grad-CAM mask-overlap analysis is explicitly interpreted as a failure-mode audit rather than localization validation.
Limitations and generalizability	Main manuscript: Discussion; Implications for applied use; Limitations	Discusses patient-level dependence, retrospective external validation, segmentation dependence, subgroup uncertainty, and non-deployment status.
Open science, reproducibility, and data availability	Main manuscript: Code and Data Availability; Reproducibility; Supplementary Methods	States fixed seed, configuration-based reproducibility, planned public code/result archive, and public dataset sources.
Ethics, funding, competing interests, and AI-assisted writing	Main manuscript: Ethics Approval; Funding; Declaration of Competing Interest; AI-assisted writing declaration	Reports public de-identified datasets, funding role, competing interests, and writing-assistance disclosure.